Calibrating Normalized Hypergraph Spectral Risk under Node-Induced Dependence

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Abstract

Prediction-constructed hypergraphs couple incident relations through shared node predictions, making edge marginals insufficient for spectral-risk assessment. We connect normalized conditional influence, calibration under a declared prediction law, and complete-run diagnostics. Conditional expectations decompose the normalized operator into orthogonal node responses and a remainder; a three-operator comparison separates fixed degree weighting from dynamic normalization. On Cora-CC and DBLP-CA, computed centered-norm ratios between node-induced and independent-edge models decreased from 2.15–3.47 before normalization to 1.03–1.26 afterward, while a coherent controlled family reversed the comparison. Under specified node laws at a 5% target, node-calibrated native exceedance means were 3.9%–5.9%, compared with 47.5%–100% for independent-edge thresholds. With trained predictors, fitted node-product thresholds instead yielded zero observed exceedances in the tested settings. This conservative direction extended to NTU2012 and ModelNet40, with zero exceedances in 200 evaluation runs per visual dataset; independent-edge calibration yielded 26.5% on ModelNet40. Complete-run and split-sensitivity comparisons distinguish marginal rank validity from the conditional behavior of a frozen threshold. A standard rank argument supplies finite-sample validity under the declared exchangeable score law. Together, the analysis and cross-domain comparisons identify which joint dependence a calibration model must preserve and separate the probability targets needed for upstream operator monitoring.

Keywords: Hypergraph learning, Relation estimation, Node influence, Spectral fluctuation, Dependent errors, Risk calibration

1. Introduction

Hypergraph learning aggregates information through group relations [1, 2]. When predictions determine which candidate relations are retained, one node prediction can change several incident hyperedges at once. An independent-edge model can match every retention probability while missing this joint variation. Risk assessment must therefore specify how the predictions generating an operator vary.

Degree normalization introduces a second distinction. Matching the mean raw relation operator does not generally match the mean normalized operator, and raw fluctuation amplification need not survive normalization. A third distinction arises with trained predictors: sampling nodes independently from their estimated class frequencies preserves their marginals but removes dependence induced by a shared training run. These distinctions concern the same question: *how does*

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node-induced dependence change normalized spectral risk, and which prediction law must calibration preserve?

We answer this question by connecting normalized conditional influence, law-specific calibration, and repeated-run diagnostics. Conditional expectations decompose the normalized operator into orthogonal single-node influences and a remainder. A three-operator comparison then distinguishes static degree weighting from dynamic normalization. Keeping this mechanism analysis separate from monitoring permits a standard finite-sample rank threshold with a precise probability target. Finally, complete training runs test whether fitted node marginals provide an adequate replacement law.

The contribution is this connection and its measured limits in prediction-constructed hypergraphs. The conditional-expectation decomposition and rank rule are established tools. Their application shows why matching edge marginals, retaining node-induced dependence, and retaining an entire training-run law are different requirements. Experiments on controlled families, Cora-CC, and DBLP-CA evaluate these requirements separately. Raw-operator amplification, cancellation, and tail-risk results supply the mechanism foundation in Appendices A and B; the main analysis asks what remains true after normalization and calibration.

Finally, we extend the complete-run calibration protocol to two visual object hypergraphs, NTU2012 and ModelNet40. These experiments test the cross-domain behavior of fitted node-product and independent-edge calibration, and distinguish marginal rank validity from the conditional variability of a threshold estimated from finitely many complete runs.

1.1. Related work

Matrix concentration controls independent matrix sums [3] and matrix functions of independent variables [4]. Other dependence assumptions include strong negative dependence and Markovian summands [5, 6]. Independence of node coordinates does not imply independence of their induced edge events. Our raw-operator comparison uses the classical ANOVA decomposition [7], related to overlap calculations for motif counts [8]; its normalized version retains the conditional-expectation identity while changing the influences. Decoupling [9], latent-position spectral inference [10], and hypergraph spectral similarity [11] address related operator questions under their respective probability models.

Rank calibration uses exchangeability to control a new observation jointly with the random calibration sample. This differs from requiring the conditional error probability of every realized calibration sample to equal a target [12]. Conformal risk control extends split-conformal reasoning to expected monotone losses [13]. We use the simpler scalar upper-tail rank rule and prove its scope explicitly. The inference target is an operator-deviation alarm under the declared law. Task-level performance is a separate empirical outcome.

Structural-noise analysis and hyperedge refinement connect graph reliability to learning performance [14, 15]. Here the learning task motivates the relation construction, and spectral deviation under a stated law serves as an upstream structural diagnostic. Its task-level consequences require separate evaluation.

2. Operators, Centers, and Prediction Laws

Fix a candidate hypergraph E on n nodes, with $|e| \geq 2$. For a prediction vector Y, define

$$\begin{aligned} H_e &= \mathbf{1}_e \mathbf{1}_e^\top / |e|, \qquad & a_e(Y) &= \mathbf{1}\{Y_u \text{ agree over } u \in e\}, \\ B(a) &= \sum_e a_e H_e, & A(a) &= B(a) + \kappa I, \\ D(a) &= \operatorname{diag}(A(a)\mathbf{1}), & S(a) &= D(a)^{-1/2} A(a) D(a)^{-1/2}. \end{aligned} \tag{1}$$

The experiments use $\kappa = 1$ and unit edge weights. Degrees are recomputed for every draw: $d_v(a) = \kappa + \sum_{e \ni v} a_e$. Thus $S(a)\sqrt{d(a)} = \sqrt{d(a)}$ and $\|S(a)\|_2 = 1$. This is the specified nonnegative symmetric operator; an implementation using another HGNN normalization requires its own definition.

Under an independent categorical node law $P[u, c]$, model A draws the entire Y and constructs every $a_e(Y)$. Model B draws edge slots independently with probabilities $p_e = \sum_c \prod_{u \in e} P[u, c]$. Duplicate node sets share an event in A but are independent slots in this counterfactual B model. Both arms have $\mathbb{E}B = \sum_e p_e H_e$. For the fixed-degree control,

$$D_0 = \operatorname{diag}((\mathbb{E}B + \kappa I)\mathbf{1}), \qquad S_0(a) = D_0^{-1/2}(B(a) + \kappa I)D_0^{-1/2}, \tag{2}$$

the means still match by linearity. Generally $\mathbb{E}_A S \neq \mathbb{E}_B S$, and $S(p) \neq \mathbb{E}_A S$. We report their roles separately.

Mechanism and monitoring centers. Mechanism comparisons use each arm's own mean in $\|S - \mathbb{E}_{\text{arm}} S\|_2$, with independently estimated means in the simulations. Monitoring uses one center $\widehat{M}$ estimated in an independent reference stage and then frozen:

$$T(Y) = \|S(a(Y)) - \widehat{M}\|_2. \tag{3}$$

Comparisons between calibrators use the same center and scoring pipeline. The deterministic baseline $S(p)$ is used only for local response analysis, not as an interchangeable estimate of $\mathbb{E}S$.

Three prediction-law levels. L1 specifies a known product law for node predictions and tests its induced joint relation law. The synthetic error profiles use benchmark labels to define this model; they are not estimates obtainable without labels at deployment. L2 estimates node class frequencies in an independent reference stage and resamples from their product. Its validity within that fitted law does not establish validity for the actual training process. L3 treats an entire prediction vector from one training run as one observation. Exchangeability is required across runs of the same protocol, rather than across nodes. Random labeled sets, random training partitions, and initialization with a fixed training set define separate L3 mechanisms.

Figure 1 separates two parts of the analysis. The operator chain isolates static degree weighting from dynamic degree recomputation, while each prediction law supplies complete relation draws to the same normalized scoring pipeline.

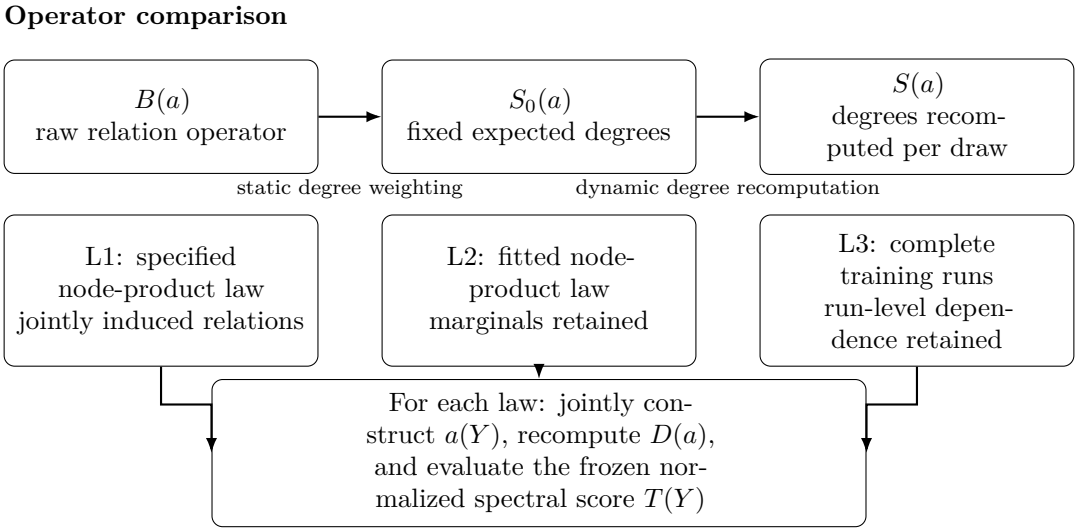


Figure 1: The operator comparison and the prediction-law comparison are distinct axes. $B \to S_0 \to S$ separates static weighting from dynamic normalization; L1–L3 specify which joint prediction mechanism generates the relation draws used by the same normalized score.

3. Normalized Node Influence

Normalization changes the conditional response, but it does not remove the conditional-expectation decomposition. Let $X^S = S(Y) - \mathbb{E}S(Y)$ under L1 and define

$$F_u^S(z) = \mathbb{E}[S(Y) \mid Y_u = z] - \mathbb{E}S(Y), \qquad R^S = X^S - \sum_u F_u^S(Y_u). \tag{4}$$

Proposition 3.1 (Normalized conditional influence)**.** *For independent node coordinates and finite second moments,* $\mathbb{E}[R^S \mid Y_u] = 0$ *for every* u*, and*

$$\mathbb{E}\|X^S\|_F^2 = \sum_u \mathbb{E}\|F_u^S(Y_u)\|_F^2 + \mathbb{E}\|R^S\|_F^2. \tag{5}$$

For a binary coordinate with $P_0P_1 > 0$*, put* $\Delta_u = \mathbb{E}[S \mid Y_u = 1] - \mathbb{E}[S \mid Y_u = 0]$*. Then* $F_u^S(1) = P_0\Delta_u$*,* $F_u^S(0) = -P_1\Delta_u$*, and* $v_u := \mathbb{E}\|F_u^S(Y_u)\|_F^2 = P_0P_1\|\Delta_u\|_F^2$*.*

Proof. Every F_u^S has mean zero. For $v \neq u$, independence gives $\mathbb{E}[F_v^S(Y_v) \mid Y_u] = 0$, so conditioning (4) on Y_u gives zero conditional remainder. Distinct influences, and each influence with the remainder, are orthogonal in the expected Frobenius inner product. Expanding the squared norm proves (5). The two binary values follow by subtracting $P_0\mathbb{E}[S \mid 0] + P_1\mathbb{E}[S \mid 1]$. □

The proposition is the classical first-order decomposition applied to the normalized matrix function. It identifies structure beyond degree: two nodes with the same incident-edge count can have different Δ_u because the remaining class profiles reinforce or cancel. It does not assert that first-order influence dominates the remainder. The raw-operator separation and cancellation theorems in Appendix A make their additional structure conditions explicit. Their exact conclusions do not transfer automatically through dynamic normalization.

The three-operator comparison makes that transfer measurable. Static weighting changes B into S_0, retaining common expectations. Dynamic degrees change S_0 into S and can change both means and fluctuations. We therefore report a centered-norm ratio and a separate mean shift, rather than interpreting a common-center difference as fluctuation alone. Local derivative channels and a general perturbation bound are given in Appendix H as supporting analysis.

4. Calibration under a Stated Prediction Law

4.1. Reference, calibration, and evaluation

The input is a fixed candidate hypergraph, its prediction-to-relation map, a specified prediction law, and a target upper-tail probability α. The output is a frozen center, a threshold, and the law under which the threshold is justified. The procedure has three stages.

1. Use independent reference observations to estimate $\widehat{M}$ and freeze all choices defining the score. For L2, estimate the product law here; for L3, an observation is a complete training run.
2. Generate M independent calibration observations from the chosen law, construct all relations jointly, recompute degrees, and obtain $T_1, \dots, T_M$ with the frozen score.
3. For evaluation, keep the center and threshold fixed. Neither new scores nor evaluation labels update the calibrator.

The edge-independent baseline replaces step 2 by independent edge-slot sampling at the same specified or fitted marginals. This changes the calibration law; it does not inherit the node-law guarantee.

4.2. Rank threshold and its probability target

Let $T_{(1)} \le \cdots \le T_{(M)}$ and define

$$k = \lceil (M+1)(1-\alpha) \rceil, \qquad q_\alpha = \begin{cases} T_{(k)}, & k \le M, \\ +\infty, & k > M. \end{cases} \tag{6}$$

An alarm is the strict event $T_{\text{new}} > q_\alpha$.

Proposition 4.1 (Law-specific rank validity)**.** *Condition on the independent reference stage. If the calibration scores and a new score are exchangeable under a law $\mathcal{L}$, then*

$$\Pr_{\mathcal{L}}\{T_{\text{new}} > q_\alpha \mid \text{reference}\} \le \alpha. \tag{7}$$

The probability averages jointly over calibration observations and the new observation. With ties, the strict rule is conservative.

Proof. Break ties by independent random ordering. Exchangeability makes the new observation's rank uniform on $1, \dots, M+1$. Strict exceedance of the kth calibration score implies a rank greater than k, giving probability at most $(M+1-k)/(M+1) \le \alpha$. Removing the random tie-breaking can only remove strict exceedances. If $k > M$ the event is empty. □

This is the standard rank argument [12]. Within L1 it applies to node draws and their jointly induced edges. Within a frozen fitted product law it applies to L2 draws from that law, not automatically to real training runs. Under L3 it applies to fresh exchangeable complete runs of the same protocol. Repeatedly resampling a finite run pool is not itself evidence of exchangeability with a fresh training run.

4.3. Interpreting empirical exceedance

The realized threshold's conditional exceedance probability need not equal α or satisfy (7) conditional on that particular calibration sample. Binomial Clopper–Pearson intervals describe evaluation variability for a frozen threshold; including α in such an interval does not prove equivalence. Across batches with different fitted centers and thresholds, pooled evaluation indicators need not have a common success probability. We report batch-specific intervals or their explicitly labeled endpoint envelope.

5. Experiments

The main experiments ask three consecutive questions: what normalization changes, whether the chosen law supports calibration, and whether a fitted node product reproduces complete training runs. Cora-CC and DBLP-CA use AllSet benchmark inputs [2]; their candidate structures contain 2,708/41,302 nodes and 1,579/22,363 edge slots. The controlled normalization families use 120 edges per hub and are distinct from the larger raw-operator grids retained in Appendix B. All numerical comparisons use the operator in (1).

Numerical and statistical protocol. Reference, calibration, and evaluation stages use separate seed streams. Small matrices use dense symmetric eigendecomposition. Larger matrices use both algebraic ends of a sparse eigensolver with a fixed start, and the implementation can return an approximation or fallback score. The rank guarantee applies to the shared implemented score, not to a certified exact spectral norm. For large-scale DBLP matrices, raw and fixed-degree spectral scores are computed with iterative sparse eigensolvers as point estimates (marked with † in Table 1), whereas rank-calibration guarantees apply directly to the shared implemented scoring pipeline.

5.1. Normalization changes the dependence comparison

Dynamic normalization attenuated the estimated centered-norm ratio on all displayed native units, while their computed ratios remained above one (Table 1). The native dynamic ratios ranged from 1.03 to 1.26, compared with raw ratios of 2.15–3.47. The coherent controlled family instead changed from 2.93 for B to 0.24 for S, accompanied by a normalized mean shift of 0.532. These outcomes separate raw amplification from normalized behavior. They do not attribute a fluctuation difference to mean shift alone, because the ratios use each arm's own estimated mean.

Structure	Profile	R_B	R_{S_0}	R_S	Δ_μ
Coherent	aligned	2.93	1.16	0.24	0.53
Competition	$\gamma = .05$	1.01	1.00	1.00	0.02
Competition	$\gamma = .50$	1.40	1.02	0.99	0.09
Competition	$\gamma = .95$	2.28	0.85	0.53	0.44
Cora	binary .05	3.25	1.39	1.24	0.08
Cora	binary .15	2.93	1.41	1.09	0.18
Cora	multi .05	3.47	1.24	1.26	0.08
Cora	multi .15	3.28	1.29	1.11	0.17
Cora	hetero	3.23	1.45	1.05	0.23
DBLP†	binary .05	2.48	1.70	1.13	0.19
DBLP†	binary .15	2.16	1.42	1.03	0.29
DBLP†	hetero	2.15	1.77	1.07	0.34

Table 1: Computed mean centered-norm ratios A/B; $\Delta_\mu = \|\widehat{\mathbb{E}}_A S - \widehat{\mathbb{E}}_B S\|_2$. Dynamic norms use each arm's own estimated mean; raw and fixed-degree means match analytically. Center/evaluation budgets per arm are 800/600 for the controlled families and Cora, and 120/80 for DBLP. Native binary/multiclass profiles have the displayed error probability; hetero uses binary errors at .02 below the upper degree decile and .30 within it. † DBLP raw/fixed-control scores include numerical approximations; these are point estimates, not certified norm ratios.

Exact small-graph enumeration verified the normalized conditional orthogonality and energy identity to numerical precision. On the degree-120 controlled hubs, estimated v_u was 1.099 for the coherent profile, approximately zero at the competing flat profile, and 1.311 at the aligned competing-family endpoint. Thus equal degree did not fix normalized influence energy. Native influence estimates used selected high-impact and background nodes; no full-native energy closure is claimed.

5.2. Model-law calibration and separate-seed reanalysis

Preserving node-induced relations sharply changed calibration under known node laws (Table 2). At target $\alpha = 5\%$, native node-calibrated batch means ranged from 3.9% to 5.9%, while independent-edge thresholds yielded 47.5%–100% exceedance under the same node mechanism. These are model-law comparisons: the labels used to define the artificial errors are known to the simulator. Heterogeneous calibration profiles use multiclass errors; the heterogeneous normalization profiles in Table 1 instead use binary errors.

Structure	Profile	Node (%)	Edge-iid (%)
Cora	binary .05	5.2 [3.7,7.2]	99.5 [98.3,100.0]
Cora	binary .15	5.0 [3.6,6.7]	97.7 [95.6,99.4]
Cora	hetero	5.9 [3.0,9.3]	47.5 [40.5,54.9]
Cora	multi .05	4.8 [3.3,6.5]	99.7 [98.7,100.0]
Cora	multi .15	5.5 [2.9,8.3]	99.0 [97.5,99.9]
DBLP	binary .05	4.6 [1.4,9.3]	100.0 [98.8,100.0]
DBLP	binary .15	5.1 [1.4,10.1]	96.8 [90.7,99.8]
DBLP	hetero	3.9 [0.9,8.9]	99.7 [97.1,100.0]
Coherent	aligned	0.0 [0.0,0.2]	9.2 [7.8,10.7]

Table 2: Known-node-law exceedance at $\alpha = 5\%$. Native entries are means over three independent threshold batches; brackets give the minimum lower and maximum upper endpoint of their individual 95% binomial intervals, not a pooled 95% interval. Per-batch reference/calibration/evaluation budgets are 200/500/1000 on Cora and 80/150/300 on DBLP. Hetero here uses multiclass errors at .02/.30. The coherent-family row is one batch with 400/800/1600 observations.

The controlled coherent family illustrates conservative ties. Its 5% node threshold had zero strict exceedances in 1,600 evaluations, because a discrete upper-tail atom determined the threshold. Zero exceedance does not establish useful detection power, even when rank validity holds. In the fitted-law Cora pilot, the three-repeat mean 5% exceedance was approximately 5.0% within the fitted model and 4.9% under the specified generating law. That observation concerns a weak-error synthetic product law, not the training-induced L3 law.

A fresh three-batch Cora rerun produced rates of 6.9%, 3.3%, and 4.9%, for a mean of 5.03% across 3,000 evaluations (Table 3). The batch-specific 95% intervals have an endpoint envelope of 2.28%–8.65%. Because centers and thresholds differ by batch, this envelope is not a pooled interval. The three-batch mean exceedance rate was 5.03%, consistent with the nominal 5% target at this resolution.

Batch	Exceedances	Rate (%)	CP 95% interval	$q_{.05}$
1	69/1000	6.9	[5.4,8.7]	0.790
2	33/1000	3.3	[2.3,4.6]	0.764
3	49/1000	4.9	[3.6,6.4]	0.782

Table 3: Fresh Cora-CC rerun under the specified binary node-product law. Each batch uses 200/500/1000 reference/calibration/evaluation draws and a distinct deterministic seed. Intervals are batch-specific conditional Clopper–Pearson intervals; the endpoint envelope is not a pooled interval. The three-batch mean exceedance rate was 5.03%, consistent with the nominal 5% target at this resolution.

5.3. Complete training runs expose product-law mismatch

The product of empirical node frequencies produced zero observed exceedances in every displayed L3 setting (Table 4). Whole-run thresholds instead produced nonzero exceedances with broad

conditional intervals. In the two fixed-training-set GCN settings, the run-pool counts at 5% were 1/60 on Cora and 1/40 on DBLP; the corresponding node-product counts were zero. These observations support using complete runs as the practical calibration baseline for the tested mechanisms. They do not establish uniqueness or universal superiority of this baseline.

Data	Mechanism	R/C/E	Run-pool (%)	Node-fit	Edge-iid
Cora	LP	50/100/100	4.0 [1.1,9.9]	0.0	0.0
DBLP	LP	40/80/80	6.2 [2.1,14.0]	0.0	0.0
Cora	Proto	30/100/150	3.3 [1.1,7.6]	0.0	9.3
DBLP	Proto	20/50/80	6.2 [2.1,14.0]	0.0	0.0
Cora	GCN-XF	20/40/60	6.7 [1.8,16.2]	0.0	0.0
Cora	GCN-init	20/40/60	1.7 [0.0,8.9]	0.0	0.0
DBLP	GCN-init	15/25/40	2.5 [0.1,13.2]	0.0	0.0

Table 4: Whole-run evaluation at $\alpha = 5\%$. R/C/E are the reference/calibration/evaluation run counts. All rates are percentages; run-pool brackets are conditional 95% binomial intervals for the realized threshold. LP varies labeled subsets, Proto varies a half split, GCN-XF uses five-fold cross-fitting, and GCN-init holds a 20% labeled subset fixed. Zero node-fit counts do not imply zero population probability; their upper 95% endpoints range from 2.4% to 8.8%. Every 5% threshold displayed is finite.

The random mechanisms differ deliberately. LP varies the labeled subset; prototype prediction varies a half split; cross-fit GCN varies fold assignments and initialization. The final two rows hold a 20% labeled subset fixed and vary initialization only, using 500 epochs on Cora and 120 on DBLP. The calibration/evaluation separation is at the run level in each case. No DBLP cross-fit GCN row is inferred from the fixed-split experiment.

At $\alpha = 10\%$, the fixed-split run-pool counts were 9/60 on Cora and 1/40 on DBLP. Their intervals included the target but were too wide to establish equivalence. At $\alpha = 1\%$, calibration pools of 25, 40, 50, or 80 runs cannot yield a finite threshold under (6); zero alarms at $q = \infty$ convey no detection ability. More simulation from a fitted node product cannot repair missing dependence in the model being simulated.

5.4. Cross-domain calibration on visual object hypergraphs

We extended the complete-run protocol to two preprocessed visual object hypergraphs from the ED-HNN benchmark [16]: NTU2012 (2,012 objects, 67 classes, 2,012 five-object hyperedges) and ModelNet40 (12,311 objects, 40 classes, 12,311 five-object hyperedges). Object features and labels were matched to the released incidence data by identifier, excluding auxiliary hyperedge-node rows. A two-layer fixed propagation predictor $S_{\text{all}}\text{ReLU}(S_{\text{all}}XW_1)W_2$ used a fixed stratified 60/20/20 training/validation/test split. Its output generated the monitored operator $S(Y)$. Each dataset supplied one batch of 30 reference, 99 calibration and 200 independent evaluation runs, for 658 complete training runs across the two datasets.

The fitted node-product calibration remained conservative across these visual domains (Table 5). At $\alpha = 5\%$, it yielded zero exceedances in 200 evaluation runs on each dataset (upper CP endpoint 1.83%), extending the direction observed on Cora-CC and DBLP-CA. Independent-edge calibration instead yielded 10/200 exceedances on NTU and 53/200 (26.5%) on ModelNet. Thus preserving edge marginals or fitted node marginals did not reproduce the calibration behavior of the complete-run law.

Complete-run calibration also exposes the distinction between marginal and conditional validity. Its frozen threshold yielded 28/200 (14.0%) exceedances on NTU and 11/200 (5.5%) on ModelNet. Proposition 4.1 averages over the calibration sample and the next exchangeable run; it does not

Dataset	Clean exceedance (%; 95% CP interval)			Resplit	Time (s)
	L3	L2	Edge-iid	mean (%)	train/score
NTU2012	14.0 [9.5,19.6]	0.0 [0.0,1.8]	5.0 [2.4,9.0]	5.0	3.5/0.04
ModelNet40	5.5 [2.8,9.6]	0.0 [0.0,1.8]	26.5 [20.5,33.2]	5.0	56.2/0.26

Table 5: Cross-domain calibration at $\alpha = 5\%$. Each dataset uses 30 reference, 99 calibration and 200 independent evaluation complete runs. L3 calibrates on complete runs, L2 on a fitted node-product law, and Edge-iid on independently sampled edge events. Brackets are batch-specific 95% Clopper–Pearson intervals. Resplit means use 10,000 random 99/200 partitions of the same 299 scores with a frozen reference center; they are internal sensitivity summaries, not independent validation. Timing gives median training/spectral-scoring seconds per complete run under the recorded six-way parallel setup.

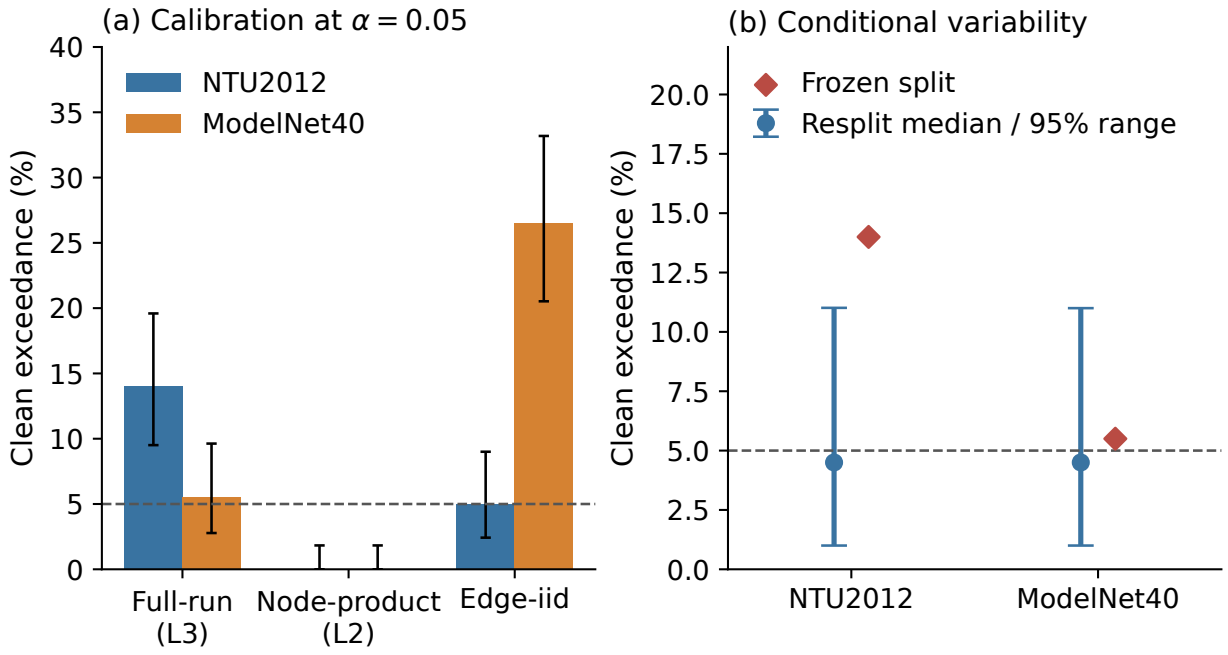


Figure 2: Cross-domain calibration and frozen-threshold variability. (a) Clean exceedance for complete-run (L3), fitted node-product (L2), and independent-edge calibration at $\alpha = 5\%$; error bars are batch-specific 95% Clopper–Pearson intervals from 200 evaluation runs. (b) Original frozen-split rates and the median and central 95% range of 10,000 random 99/200 splits of the same 299 scores, conditional on the frozen reference center. The split range is an internal sensitivity summary, not a confidence interval for the population risk. Dashed lines mark the 5% target.

require each frozen threshold's conditional exceedance probability to equal 5%. Figure 2 compares the observed rates with internal split sensitivity. With the reference center frozen, 10,000 random 99/200 splits of the same 299 NTU scores gave mean exceedance 5.0%, median 4.5%, and 97.5th percentile 11.0%; the original threshold lay at the 0.4th percentile of resampled thresholds. ModelNet also gave a split mean of 5.0%. This finite-pool rank check illustrates conditional variability; repeated use of the same runs is not an independent population-level test of the marginal guarantee.

A deployment requirement on the conditional risk of a frozen threshold calls for an additional calibration criterion. Under an independently and identically distributed run mechanism, a binomial tolerance rule controls the probability that this conditional risk exceeds α, with confidence parameter $1 - \delta$ [12]. At $M = 99$ and $\alpha = \delta = 5\%$, it uses the 98th rather than the 95th ordered score. On the same evaluation runs this gives 19/200 (9.5%) on NTU and 3/200 (1.5%) on ModelNet. These realized rates remain distinct from the rule's probability statement over calibration samples. The

tolerance comparison reuses the observed evaluation data; the original thresholds and counts are retained. Appendix H.2 gives the criterion, numerical sensitivity, and secondary feature-perturbation comparisons.

6. Discussion and Limitations

The results locate two distinct failures of a calibration shortcut. Within a known independent-node model, treating induced edge events as independent can produce severe over-exceedance. With real training runs, a product of fitted node marginals can instead yield an overly conservative threshold. The direction of failure depends on which joint law is omitted. Consequently, preserving shared-node dependence recovers the specified node-model construction but does not by itself recover a training-run distribution.

Normalization explains why the raw theory is a starting point rather than a sufficient calibration model. It can attenuate fluctuations, shift expectations, or change the direction of a controlled comparison. Conditional influence quantifies this change without assuming the remainder is small. Its energy is structural information, not a proved optimal monitoring direction. Appendix H gives a local-response formula and perturbation bound as supporting tools.

The guarantees retain explicit boundaries. Known node laws need not describe shared training states; fitted frequencies need not represent repeated-run probabilities; and a fixed data set and training protocol do not cover a new graph or a changed training process. The Cora-CC/DBLP-CA L3 evaluation pools are small and lack a new prospective confirmation with retained run-level prediction caches. The visual protocol retains run-level caches for two additional datasets under its stated single-batch 30/99/200 design. All runs in a calibration claim must remain exchangeable under its declared mechanism.

The visual benchmarks extend the distinction between joint laws to a different data domain. Fitted node products remained conservative on both, while independent-edge calibration gave substantial exceedance on ModelNet40. The NTU frozen threshold and split-sensitivity analysis further separate the probability target from any one realization: marginal rank validity does not give a high-confidence bound on the conditional risk of every deployed threshold. Binomial tolerance calibration provides an established way to express the latter objective under an independent-run model. More calibration runs can reduce sampling variation, but the probability target and assumed law must first be specified.

Numerical precision is a separate restriction. The DBLP entries in Table 1 are computed-score evidence; certified spectral-norm magnitudes require an independent numerical audit. The ES importance estimates also do not provide a usable tail certificate: their effective-sample diagnostics were low and tested bounds were vacuous. Stronger remainder bounds and risk-weighted propagation are outside the present claim.

7. Conclusion

Normalized spectral-risk calibration requires a declared joint prediction law. Conditional influence separates the effects of normalization from raw-operator fluctuation, while rank calibration states validity under the corresponding exchangeability assumption. Controlled families, citation networks and visual object hypergraphs show different calibration consequences of omitting shared-node relations or training-run dependence. Complete-run and split-sensitivity comparisons further distinguish marginal validity from the conditional behavior of a frozen threshold. These results guide the construction and interpretation of upstream operator-deviation alarms.

Data and Code Availability

The complete source code, frozen configurations, certificates, saved measurements, and reproducibility instructions for both the normalized calibration experiments and the raw-operator mechanism checks are publicly available at
`https://github.com/pengshen1212/hypergraph-spectral-fluctuations`. The normalized calibration materials are also provided in the accompanying submission package. Frozen visual prediction caches are available separately as a GitHub Release attachment, with per-file SHA-256 values in the repository's cache manifest. Cache-based reanalysis also requires the original third-party inputs. Cora-CC and DBLP-CA inputs are available through AllSet; preprocessed NTU2012 and ModelNet40 visual hypergraphs are obtained from the public ED-HNN benchmark repository [16]. The retained raw Walmart experiments use Austin R. Benson's dataset collection. Third-party inputs are not redistributed in the revision package.

CRediT Authorship Contribution Statement

Shen Peng: Conceptualization, Methodology, Formal analysis, Validation, Software, Investigation, Data curation. **Rui Wang:** Investigation, Data curation. **Yunchao He:** Writing – original draft, Writing – review and editing.

Funding

This work received no specific funding from public, commercial, or non-profit funding agencies.

Declaration of Competing Interest

The authors declare that they have no competing interests.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

ChatGPT (OpenAI) was used for literature discovery, assistance with intermediate mathematical derivations, and language revision. Codex (OpenAI) assisted manuscript organization, reference verification, language revision, and preparation of the explanatory diagram's source. The authors are responsible for checking the derivations, references, and final manuscript and for the content of the published article.

Acknowledgements

The authors thank Jianjia Wang for guidance during this research.

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Appendix A. Raw-Operator Model and Mechanism

The following raw-operator theory supplies the mechanism foundation. Its matched expectations and separation claims concern the unnormalized operator; the normalized analysis in Section 3 does not assume that these conclusions transfer.

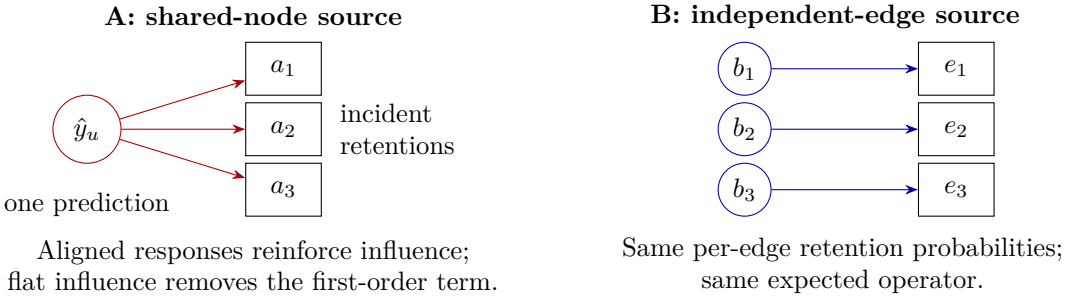


Figure A.1: Shared predictions couple incident relations even when each edge has the same marginal retention probability as an independent draw. Coherent influence measures their combined response. Norm separation also requires the stated overlap conditions; exact directional-variance cancellation requires every shared node to be flat and edge pairs to share at most one node. The explanatory diagram was drafted with Codex (OpenAI) assistance and rendered from its TikZ source.

Fix a candidate hypergraph E with edge sizes $r_e \geq 2$ and true labels y. Node predictions $\hat{y}_u$ are independent with categorical laws $P[u, c]$. These laws define the controlled relation-construction model; Appendix E describes conditioning on a shared training state. Write

$$H_e = \mathbf{1}_e \mathbf{1}_e^\top / r_e, \qquad a_e = \mathbf{1}\{\text{all predictions on } e \text{ agree}\}, \qquad p_e = \mathbb{E}\, a_e = \sum_c \prod_{u \in e} P[u, c].$$

The estimated relation operator is $B(\hat{a}) = \sum_e a_e H_e$, an unnormalized sum of rank-one clique contributions. With the true-label indicator a_e^*, the error operator is $\Delta B_A = \sum_e (a_e - a_e^*) H_e$ and $\mathbb{E}\Delta B_A = \sum_e (p_e - a_e^*) H_e =: M_0$. Both arms below are centered on this common mean, so all statements concern

$$X_A = \sum_e (a_e - p_e) H_e, \qquad X_B = \sum_e (b_e - p_e) H_e, \qquad b_e \sim \text{Bern}(p_e) \text{ independently},$$

with $\widetilde{\beta}_M = \|X_M\|_2$. The conditional retention kernel

$$g_{e,u}(z) := \mathbb{E}[a_e \mid \hat{y}_u = z] = \prod_{v \in e \setminus \{u\}} P[v, z]$$

is defined by re-multiplying over the remaining nodes, so zero probability classes are handled without division. For binary laws set $\phi_{u,e} := g_{e,u}(1) - g_{e,u}(0)$ and, for a fixed unit vector x,

$$h_e(x) := x^\top H_e x = \left(\sum_{u \in e} x_u\right)^2 / r_e \;\geq 0, \qquad \Phi_u(x) := \sum_{e \ni u} \phi_{u,e}\, h_e(x).$$

$\Phi_u(x)$ is node u's *coherent influence* at x: the amount by which its prediction moves the incident weighted retention sum.

The homogeneous submodel has all candidate edges r-uniform and homogeneous, i.i.d. binary flips $W_u \sim \text{Bernoulli}(p)$, $q_r = 1 - (1-p)^r - p^r$. There, $g_{e,u}(1) = (1-p)^{r-1}$, $g_{e,u}(0) = p^{r-1}$, and $\phi_{u,e} = \pm\left[(1-p)^{r-1} - p^{r-1}\right]$ for every member of a homogeneous edge. The general influence theory allows heterogeneous categorical laws.

A.1. Node-Influence Decomposition

The decomposition, its second-moment identity, and the variance lower bound form one chain and are stated together.

Theorem A.1 (Influence decomposition and variance identity)**.** *Define* $F_u(z) := \sum_{e \ni u} \big[g_{e,u}(z) - p_e\big] H_e$*. Then:*

(i) $X_A = \sum_u F_u(\hat{y}_u) + R$ *with* $\mathbb{E}[R \mid \hat{y}_u] = 0$ *for every* u*; moreover* $\mathbb{E}F_u(\hat{y}_u) = 0$*, the components are entrywise orthogonal (*$\mathbb{E}[F_u(\hat{y}_u)F_v(\hat{y}_v)] = 0$ *and* $\mathbb{E}[F_u(\hat{y}_u)R] = 0$ *as matrices for* $u \neq v$*), and*

$$\mathbb{E}[X_A^2] = \sum_u \mathbb{E}\big[F_u(\hat{y}_u)^2\big] + \mathbb{E}[R^2], \qquad \mathbb{E}[X_B^2] = \sum_e p_e(1-p_e)H_e.$$

(ii) *For every fixed unit* x*,*

$$\mathrm{Var}(x^\top X_A x) = V_1(x) + \mathrm{Var}(x^\top R x) \;\geq\; V_1(x), \qquad V_1(x) = \sum_u \mathrm{Var}_{Z \sim P[u,\cdot]}\Big[\sum_{e \ni u} g_{e,u}(Z) h_e(x)\Big],$$

and in the binary case $V_1(x) = \sum_u P[u,1]P[u,0]\,\Phi_u(x)^2$ *with the single-term bound* $V_1(x) \geq m_u(1-m_u)\Phi_u(x)^2$*,* $m_u = \min_c P[u,c]$*.*

Proposition A.2 (Covariance form of the A/B gap)**.** $\mathbb{E}[X_A^2] - \mathbb{E}[X_B^2] = \sum_{e \neq f} \mathrm{Cov}(a_e, a_f)\, H_e H_f$*, a symmetric operator identity;* $\mathrm{Cov}(a_e, a_f) = 0$ *whenever* $e \cap f = \varnothing$*, and for a pair sharing exactly one node* u*,* $\mathrm{Cov}(a_e, a_f) = \mathbb{E}_{Z \sim P[u,\cdot]}\big[g_{e,u}(Z) g_{f,u}(Z)\big] - p_e p_f$*. For every fixed unit* x*,*

$$\mathrm{Var}(x^\top X_A x) - \mathrm{Var}(x^\top X_B x) = 2 \sum_{e<f,\, e \cap f \neq \varnothing} \mathrm{Cov}(a_e, a_f)\, h_e h_f.$$

The proofs (Appendix C) use $F_u(\hat{y}_u) = \mathbb{E}[X_A \mid \hat{y}_u]$ per Theorem A.1, and orthogonality follows from independence and zero means. No structure assumption is made on E—sizes, overlaps, and label compositions are arbitrary. The classical star variance amplification is the special case $x = \mathbf{e}_v$ at a disjoint-leaf star: $\Phi_v = d\big[(1-p)^{r-1} - p^{r-1}\big]/r$ gives $\mathrm{Var}_A \,/\, \mathrm{Var}_B = 1 + (d-1)\kappa$ with $\kappa = p(1-p)\big[(1-p)^{r-1} - p^{r-1}\big]^2/(q_r(1-q_r))$; e.g. $(r,p,d) = (3, 0.05, 20)$ gives 6.98.

Lemma A.3 (Deterministic single-node deviation)**.** *In the binary case* $x^\top F_u(\hat{y}_u)x = \psi_u$ *takes exactly two values,* $\psi_u(1) = P[u,0]\,\Phi_u(x)$ *and* $\psi_u(0) = -P[u,1]\,\Phi_u(x)$*; hence* $|\psi_u| \geq m_u|\Phi_u(x)|$ *surely.*

A.2. Influence Separation and Its Boundary

A.2.1. Certified separation under conditioned influence

The norm-level statement needs one structural quantity beyond Φ_u. For a fixed direction x let $L_u(x) := \sum_{e \ni u} h_e(x)$ (only edges containing the witness node contribute when x is supported on it).

Theorem A.4 (Influence separation, conditioned)**.** *Binary laws. Let* $B_1, \ldots, B_t$ *be pairwise node-disjoint blocks, each containing a hub* u_i *with amplifying value* z_i *(*$m_i = P[u_i, z_i] \leq \frac{1}{2}$*), and assume* no edge contains two hubs*. Fix witnesses* $x_i = \mathbf{e}_{u_i}$ *and assume, for every* i*: (i)* coherent influence*: with* $\Phi_i := \sum_{e \ni u_i} \phi_{u_i,e} h_e(x_i)$ *(all edges containing* u_i *contribute, whatever block or environment they belong to),* $|\Phi_i| \geq B$*; (ii)* incoherent rest*:* $V^- = \max_i \sum_{u \neq u_i} P[u,1]P[u,0]\Phi_u(x_i)^2$ *and* $\phi^-_{\max} = \max_{i, u \neq u_i} |\Phi_u(x_i)|$*; (iii)* bounded co-occurrence*:* $\Lambda^2 = \max_i \sum_{u \neq u_i} L_u(x_i)^2$*. Then*

$$\Pr\Big\{\widetilde{\beta}_A \;\geq\; L_A^{\inf}\Big\} \;\geq\; 1 - \prod_i (1-m_i) - \eta - \eta_2,$$

with

$$L_A^{\inf} = \min_i (1 - m_i)\, B - \sqrt{2V^- \log(2/\eta)} - \tfrac{\phi^-_{\max}}{3} \log(2/\eta) - \sqrt{2\Lambda^2 \log(2/\eta_2)}.$$

The B arm satisfies, with probability $\geq 1 - \delta_B$, $\widetilde{\beta}_B \leq U_B = \sqrt{2 v_B L_n} + 2L_n/3$ *with* $v_B = \left\| \sum_e p_e(1 - p_e) H_e \right\|_2$, $L_n = \log(2n/\delta_B)$ *(the conservative Bernstein form; the tight* $+L_n/3$ *inversion also holds); if the deterministic condition* $L_A^{\inf} > U_B$ *holds, then* $\Pr\{\widetilde{\beta}_A > \widetilde{\beta}_B\} \geq 1 - \prod_i (1 - m_i) - \eta - \eta_2 - \delta_B$.

Edges may have mixed labels, arbitrary sizes, and shared co-members; their witness co-occurrence mass enters Λ^2. At the coordinate witness x_i, only edges containing u_i contribute. The one-hub-per-edge condition keeps conditioned means unchanged by other hubs. All witnesses are fixed before sampling. For homogeneous stars the leading constant is $(1-p)d[(1-p)^{r-1} - p^{r-1}]/r$; cyclic overlap enters through Λ^2. Their verification grids are in Appendix G. The complementary budget certificate for a star core with an overlapping environment is in Appendix D.

Corollary A.5 (Scale separation)**.** *Binary laws, arbitrary overlap. If* $|\Phi_{u^*}(x)| \geq B$ *at a unit* x*, then with* $m = \min_c P[u^*, c]$, $\theta \in (0,1)$*, and* $Z_{\max} := \sum_e h_e(x) \geq |x^\top X_A x|$*,*

$$\Pr\Big\{ |x^\top X_A x| \;\geq\; \sqrt{\theta\, m(1-m)}\, B \Big\} \;\geq\; (1-\theta)^2\, \frac{m(1-m)B^2}{Z_{\max}^2},$$

by Paley–Zygmund applied to Theorem A.1(ii) (proof in Appendix C); when the influence is coherent ($Z_{\max} = O(B)$) *this is a constant-probability, degree-linear bound on arbitrary overlapping structures, against* $\widetilde{\beta}_B = O(\sqrt{v_B \log n})$ *with high probability.*

A.2.2. Cancellation boundary

Theorem A.6 (Flat profiles and exact cancellation)**.** *Binary laws. (i)* $\phi_{u,e} = 0$ *iff* u*'s prediction does not change the conditional retention of* e*; a flat edge drops out of* Φ_u *for every* x*, and if all of* u*'s edges are flat then* u *contributes nothing to* $V_1(x)$ *for any* x*. (ii) For a size-*3 *edge, if the two members* w, w' *of* $e \setminus \{u\}$ *satisfy* $P[w,1] = 1 - P[w',1]$ *(a competing pair), then* $\phi_{u,e} = 0$*: on a triple, one competing pair zeroes the third member's influence. (iii) If* $r_e \geq 3$*, all members of* e *are flat, and every member has* $P[u,1]P[u,0] > 0$*, then all members' laws are uniform on the two classes. (iv) If distinct edges share at most one node and every shared node is flat on both incident edges, then* $\mathrm{Cov}(a_e, a_f) = 0$ *for all* $e \neq f$*, hence* $\mathrm{Var}(x^\top X_A x) = \mathrm{Var}(x^\top X_B x)$ for every x*: second-order cancellation is exact.*

The homogeneous-star case $p = \frac{1}{2}$ is a symmetric instance of flatness. A one-parameter knob makes the transition explicit: for edges $\{u, w, \ell\}$ with $P[w,1] = \beta$ and $P[\ell,1] = \gamma$,

$$\phi_{u,e} = \beta + \gamma - 1,$$

so sweeping γ from alignment ($\beta = \gamma$) to competition ($\gamma = 1 - \beta$) drives the hub's influence linearly to the exact cancellation endpoint of part (iv). Section B.1 sweeps this knob.

Higher-order dependence and norm-level separation. The cancellation theorem identifies directional second moments. Spectral-norm distributions also depend on higher-order terms in R. For edge pairs sharing multiple nodes, these terms can retain pair covariance despite flat single-node profiles. Norm separation is certified when coherent drift exceeds the remainder and co-occurrence penalties.

A.2.3. Conditional influence and tail-risk certification

The same conditioning idea permits a native-structure risk judgment without the overlap restrictions of Theorem A.4. Fix a small node group U from structure alone and $\tau \geq 0$. For each positive-probability state z, let $\pi_z = \Pr(\hat{y}_U = z)$ and $M_U(z) = \mathbb{E}[X_A \mid \hat{y}_U = z]$. A nonzero direction q_z is chosen from this conditional mean before evaluation; it may depend on z, since the states are disjoint. Write

$$s_z = q_z^\top q_z, \qquad m_z = q_z^\top M_U(z) q_z, \qquad v_z = \mathrm{Var}(q_z^\top X_A q_z \mid z), \qquad D_z = |m_z| - \tau s_z.$$

Rayleigh's inequality and one-sided Cantelli give

$$\Pr(\|X_A\|_2 > \tau) \;\geq\; L_U(\tau) := \sum_z \pi_z \ell_z(\tau), \qquad \ell_z(\tau) = \begin{cases} D_z^2/(v_z + D_z^2), & D_z > 0, \\ 0, & D_z \leq 0. \end{cases} \tag{A.1}$$

The bound uses exact conditional covariances, including overlaps outside U. Appendix F gives the calculation and calibration guarantee. If τ_B controls B's tail at level α and $L_U(\tau_B) > \alpha$, dependence is certified to invalidate that threshold for A. When $L_U(\tau_B) \leq \alpha$, the test is inconclusive. Calibrating a replacement threshold requires a tail upper bound or quantile analysis.

Appendix B. Raw-Operator Validation and Predictor Comparisons

These retained measurements concern the raw relation operator and the original per-edge calibration protocol. They are distinct from the normalized calibration experiments in Section 5. The Walmart inputs use the categorical benchmark associated with [17]. All original result tables are retained below.

B.1. Controlled structure families

The controlled families separate the role of influence from topology (Figure A.1). Aligned responses amplify fluctuation on non-star structures; changing only the probability profile can remove that amplification on a fixed star. Both registered families use 256 formal draws per point. Family A (*amplification*): $t = 60$ node-disjoint blocks, each with one weak hub ($P[u_i, 1] = 0.05$) and d edges $\{u_i, w_j, \text{leaf}\}$ whose co-members are strongly class-1 ($P = 0.95$), with w_j drawn uniformly from a per-block pool of $d/5$; Λ^2 is computed from realized co-occurrences. Every edge at u_i has $\phi_{u_i,e} = 0.90$, so $\Phi_i = 0.9\, d/3$ while all other nodes are exactly flat ($V^- = 0$). Edges overlap through the pooled w_j's, and the blocks combine heterogeneous probability profiles. Table B.1 reports the grid. The mean centered A norm grows linearly in d (fitted slope 0.276 on $d \in \{150, \ldots, 300\}$, within 8% of the theoretical 0.9/3) while B grows more slowly; separation frequency is 93.8–100%, above the registered allowance $1 - 0.95^{60} - \eta - \eta_2 - \delta_B = 0.924$ throughout. With two-sided constants and realized $\Lambda^2 \in [47.6, 250.0]$, the certificate activates at $d \geq 250$ (at $d = 200$: $L_A = 14.4$ vs $U_B = 19.7$). Disjoint-star and cyclic checks support the same mechanism (Appendix G).

Family B (*competition sweep*) fixes one hub with $d = 300$ edges $\{u, w_j, \ell_j\}$, all co-members fresh. Its *topology is a star*. Only the probability profile changes: $P[u, 1] = 0.05$, $P[w_j, 1] = 0.95$, and $P[\ell_j, 1] = \gamma$, giving $\phi_{u,e} = \gamma - 0.05$. In Table B.2, influence falls linearly to zero and the mean centered-norm ratio falls from 3.36 to 0.995. At $\gamma = 0.05$, the empirical directional variance ratio is 0.96 with paired-bootstrap interval $[0.76, 1.20]$, covering the exact prediction 1. Edge pairs share only the flat hub, so Theorem A.6(iv) gives equality in every direction. The empirical separation frequency is 55.5% at this endpoint. Thus fixed star topology admits both amplification and cancellation, while certification also depends on overlap structure.

d	Φ	L_A	U_B	$\overline{\widetilde{\beta}}_A$ [CI]	$\overline{\widetilde{\beta}}_B$	sep.
50	15	−8.2	13.7	16.3 [15.8, 16.8]	2.36	93.8%
100	30	−1.7	16.2	30.9 [30.1, 31.5]	2.90	96.1%
150	45	6.0	18.1	43.9 [42.5, 45.1]	3.33	94.1%
200	60	14.4	19.7	57.4 [55.4, 59.3]	3.68	94.5%
250	75	23.75	21.0	72.5 [70.8, 74.1]	4.05	98.0%
300	90	34.0	22.3	85.4 [83.0, 87.6]	4.28	100%

Table B.1: Family A (non-star amplification, t=60, pool $d/5$ with random reuse, 256 draws per point). $\Phi = 0.9d/3$ is the hub's coherent influence; L_A uses two-sided constants and the realized co-occurrence Λ^2. Certification is nonempty from $d = 250$; empirical separation exceeds the registered allowance 0.924 everywhere.

γ	Φ_u	$\overline{\widetilde{\beta}}_A$	$\overline{\widetilde{\beta}}_B$	ratio	sep.
0.95	90.0	8.02	2.39	3.36	100%
0.75	70.0	6.38	2.47	2.59	89.8%
0.55	50.0	5.03	2.36	2.13	79.7%
0.35	30.0	3.88	2.52	1.54	48.8%
0.15	10.0	2.76	2.41	1.14	42.6%
0.05	0.00	2.39	2.40	1.00	55.5%

Table B.2: Family B competition sweep on a star topology (d=300, 256 draws per point). At $\gamma = 0.05$ the hub is exactly flat ($\phi_u = 0$); the directional variance ratio is 0.96 with CI $[0.76, 1.20]$ covering the predicted equality.

B.2. Native applicability

On unmodified Cora-CC and DBLP-CA, conditional influence certifies excess alarm risk where the original concentration certificate is empty. Independent-edge thresholds frequently flag node-driven errors on the same fixed structure, although edge marginals and the expected operator are matched. Thus the alarm difference concerns the error law, rather than an underlying structural change. The frozen error laws are majority-versus-rest binary flips and symmetric multiclass errors: $P[u, y_u] = 1 - \varepsilon$, with remaining mass $\varepsilon/(C-1)$ per other class, $\varepsilon \in \{0.05, 0.15\}$. In the binary case y_u is the binarized label. Groups consist of the maximum-degree node and its largest weighted co-edge partner; state-specific directions follow the largest absolute eigenvalue of $M_U(z)$ on its nonzero support. Group-selection ties use node IDs; all positive-probability states are retained. Exact rational moments certify the saved directions even when eigenvectors are approximate. Groups, directions, and moments are fixed before fresh calibration; each threshold-dependent certificate precedes validation.

Each cell uses 2,000 fresh B calibration draws, nominal tail 5%, and joint calibration confidence at least 99% across eight cells. Two independent 1,000-draw validation batches per arm give the alarm-rate envelopes in Table B.3. Node-driven alarms occur in 93.9–100% of validation draws, against a nominal 5% independent-edge tail. All eight exact bounds exceed 5%; all eight Bonferroni-adjusted A confidence lower endpoints exceed 92%. Calibration confidence and familywise 95% validation intervals are separate guarantees. The alarm-rate envelopes include all 49 ambiguous norm intervals. B's 5.1% endpoint is a finite-sample validation envelope; the calibration guarantee concerns the population tail.

Earlier variance diagnostics remain in Appendix F.

B.3. Real predictors and per-edge calibration

Empirical amplification also appears with trained predictors: the node-driven arm has larger mean centered norms in every displayed setting (Table B.4). Real predictions vary with training

Graph	Error law	ε	$L_U(\tau_B)$	A alarms (%)	B alarms (%)
Cora	Binary	0.05	0.0968	[95.85, 96.00]	[4.75, 4.75]
Cora	Binary	0.15	0.1485	[97.90, 98.80]	[3.10, 3.15]
Cora	Multiclass	0.05	0.0969	[93.90, 94.05]	[3.95, 3.95]
Cora	Multiclass	0.15	0.1513	[98.10, 99.00]	[3.80, 3.80]
DBLP	Binary	0.05	0.0950	[100, 100]	[3.65, 3.65]
DBLP	Binary	0.15	0.2550	[100, 100]	[4.15, 4.20]
DBLP	Multiclass	0.05	0.0970	[100, 100]	[5.00, 5.10]
DBLP	Multiclass	0.15	0.2730	[100, 100]	[3.65, 3.80]

Table B.3: Native tail-risk confirmation, 48,000 fresh draws in total. L_U reports a rounded exact-model probability lower bound. Alarm-rate envelopes pool 2,000 validation draws per arm and span the counts allowed by ambiguous norm intervals.

partitions or seed labels on fixed feature matrices and hypergraphs. An independent calibration group estimates $\hat{q}_e$, and an evaluation group is compared with B draws at these rates; both arms are centered on $\hat{M} = \sum_e s_e \hat{q}_e H_e$ (s_e the per-edge error sign). We report the ratio of mean centered norms. The retained settings span citation (Cora-CC), co-authorship (DBLP-CA), and co-purchase (Walmart-trips, LP only—no node features) structures: all displayed bootstrap lower bounds exceed one. The predictor families provide varied sources of relation-construction errors. Calibration sensitivity and three disjoint-seed batches are in Appendix G; all 24 batch intervals remain above one (smallest lower endpoint 1.016).

A node-marginal resampling control checks whether shared-node construction can reproduce the observed amplification. Drawing each node's prediction independently from its cross-fit empirical distribution reproduces the real arm's centered norm (3.80 vs 3.87 on Cora, ratio 0.98 $[0.86, 1.12]$; 2.79 vs 2.83 on DBLP, 0.99 $[0.94, 1.03]$), with both arms exceeding the independent-edge arm (≈ 1.7). On Cora, variance along the top eigenvector of $B(a^*)$ decomposes into 0.134 within-edge, 0.320 overlapping-pair, and 0.0003 disjoint-pair contributions. The resampling and variance decomposition are consistent with the shared-node amplification mechanism.

Dataset	Predictor	Accuracy	Centered A/B	95% interval
Cora	Prototype	0.69	2.359	[2.022, 3.203]
DBLP	Prototype	0.76	1.960	[1.715, 2.523]
Cora	GCN	0.67	2.16	[1.68, 2.90]
Cora	LP	0.57	2.11	[1.57, 3.17]
DBLP	LP	0.85	2.08	[1.67, 2.48]
Walmart	LP	0.64	4.00	[2.91, 7.51]

Table B.4: Centered A/B under per-edge calibration (20 calibration and 20 evaluation runs per cell; prototype rows from the saved 1000-replicate bootstrap at $N_{\text{cal}} = 20$, GCN/LP rows retained from the source runs). Pointwise intervals on fixed structures.

Secondary spectral readout. A fixed-k projector-distance comparison to the common expected operator separates on Cora but not DBLP (Appendix F). DBLP's positive but small gap at $p = 0.15$ permits a unique reference subspace that may be perturbation-sensitive. With k fixed, the subspace readout separates the arms on Cora and remains unresolved on DBLP.

Appendix C. Proofs of the Influence Theorems

Proof of Theorem A.1. Fix u and z. Given $\hat{y}_u = z$, the events $\{a_e = 1\}$ for $e \ni u$ depend on the other members through "all equal z", of probability $\prod_{v \in e \setminus \{u\}} P[v, z]$; edges without u have unchanged conditional mean. Hence $\mathbb{E}[X_A \mid \hat{y}_u = z] = F_u(z)$, i.e. $F_u(\hat{y}_u) = \mathbb{E}[X_A \mid \hat{y}_u]$ pathwise. Then for $R = X_A - \sum_v F_v(\hat{y}_v)$ and every u,

$$\mathbb{E}[R \mid \hat{y}_u] = \mathbb{E}[X_A \mid \hat{y}_u] - F_u(\hat{y}_u) - \sum_{v \neq u} \mathbb{E}\big[F_v(\hat{y}_v) \mid \hat{y}_u\big] = 0,$$

since $\hat{y}_v$ is independent of $\hat{y}_u$ and $\mathbb{E}F_v(\hat{y}_v) = \mathbb{E}\,\mathbb{E}[X_A \mid \hat{y}_v] = 0$. Orthogonality holds entrywise, which is what the matrix identity needs: for $u \neq v$ and any entries $(i,k), (k,j)$, $\mathbb{E}[(F_u)_{ik}(F_v)_{kj}] = \mathbb{E}[(F_u)_{ik}]\,\mathbb{E}[(F_v)_{kj}] = 0$ by independence (and likewise $\mathbb{E}[(F_u)_{ik}R_{kj}] = \mathbb{E}[(F_u)_{ik}\,\mathbb{E}[R_{kj} \mid \hat{y}_u]] = 0$), so $\mathbb{E}[F_u F_v] = 0$ and $\mathbb{E}[F_u R] = 0$ as matrices; in particular the Frobenius inner products vanish. The matrix identity follows by expanding $\mathbb{E}[X_A^2] = \mathbb{E}[(\sum_u F_u + R)^2]$ and collecting the zero cross terms; for X_B, the summands $(b_e - p_e)H_e$ are independent, centered, and idempotent-scaled ($H_e^2 = H_e$). □

Proof of Proposition A.2. Expand $\mathbb{E}[X_A^2]$ as

$$\sum_{e,f} \mathbb{E}[(a_e - p_e)(a_f - p_f)] H_e H_f.$$

The diagonal is $\sum_e p_e(1-p_e)H_e = \mathbb{E}[X_B^2]$, and $e \cap f = \varnothing$ makes a_e, a_f independent (functions of disjoint node sets). For $e \cap f = \{u\}$, conditionally on $\hat{y}_u$ the two events depend on disjoint node sets, so $\mathbb{E}[a_e a_f \mid \hat{y}_u] = g_{e,u}(\hat{y}_u) g_{f,u}(\hat{y}_u)$. The quadratic-form identity is the same expansion evaluated at x; $(H_e H_f)^\top = H_f H_e$ keeps the sum symmetric. □

Proof of Theorem A.1(ii) and Lemma A.3. Write $x^\top X_A x = \sum_u \psi_u + \rho$ with $\psi_u = x^\top F_u(\hat{y}_u) x$ and $\rho = x^\top R x$. The pairs (ψ_u, ψ_v) $(u \neq v)$ and (ψ_u, ρ) are uncorrelated by the orthogonality above, so the variance splits as claimed. In the binary case $\psi_u(z) = \sum_{e \ni u} (g_{e,u}(z) - p_e) h_e$ with $p_e = P[u,1] g_{e,u}(1) + P[u,0] g_{e,u}(0)$, giving $\psi_u(1) = P[u,0]\Phi_u(x)$, $\psi_u(0) = -P[u,1]\Phi_u(x)$, and $\mathrm{Var}(\psi_u) = P[u,1]P[u,0]\Phi_u(x)^2$. □

Proof of Theorem A.4. Let $E_i' = \{\hat{y}_{u_i} = z_i,\ \hat{y}_{u_j} \neq z_j\ \forall j < i\}$ be the *first realized* block events; they are disjoint with $\bigcup_i E_i' = \{\exists i : \hat{y}_{u_i} = z_i\}$, of probability $1 - \prod_i (1 - m_i)$. Fix i and work conditionally on E_i'; under this conditioning $\hat{y}_{u_i} = z_i$ and every earlier hub is pinned at its non-amplifying value, while every non-hub node's prediction keeps its unconditional law. By the no-shared-hub hypothesis, no edge at the witness x_i contains any hub other than u_i (edges without u_i have $h_e(x_i) = 0$), so pinned earlier hubs enter no term below. Decompose

$$x_i^\top X_A x_i = \underbrace{\sum_{e \ni u_i} \big(g_{e,u_i}(z_i) - p_e\big) h_e}_{=:D} + \underbrace{\sum_{u \neq u_i} \psi_u}_{=:S} + \underbrace{x_i^\top R x}_{=:\rho},$$

which is exact because $x_i^\top X_A x_i = \psi_{u_i} + \sum_{u \neq u_i} \psi_u + \rho$ by Theorem A.1 and $\psi_{u_i} = D$ on E_i'. *Drift.* From $p_e = m_i g_{e,u_i}(z_i) + (1 - m_i) g_{e,u_i}(\bar{z}_i)$,

$$g_{e,u_i}(z_i) - p_e = (1 - m_i)\big(g_{e,u_i}(z_i) - g_{e,u_i}(\bar{z}_i)\big) = \pm(1 - m_i)\phi_{u_i,e},$$

so $D = \pm(1 - m_i)\Phi_i$ and $|D| \geq (1 - m_i)B \geq \min_j (1 - m_j) B$. *First-order rest.* Every other hub has $\psi_{u_j}(x_i) \equiv 0$ (no edge contains both u_i and u_j), so S runs over non-hub nodes only: their

ψ_u are functions of the independent $\hat{y}_u$, whose laws the conditioning does not touch, and each is centered (Theorem A.1), bounded $|\psi_u| \le |\Phi_u(x_i)| \le \phi^-_{\max}$ by Lemma A.3, with variance sum $\le V^-$; a one-sided Bernstein bound applied to S and to $-S$, each at level $\eta/2$, gives $\Pr\{|S| > \sqrt{2V^- \log(2/\eta)} + \frac{\phi^-_{\max}}{3}\log(2/\eta) \mid E'_i\} \le \eta$. *Remainder.* Since $x_i^\top X_A x_i = \sum_{e \ni u_i}(a_e - p_e)h_e$, we can write $\rho = W - S$ with $W = \sum_{e \ni u_i}(a_e - g_{e,u_i}(z_i))h_e$, which depends only on the non-hub members of u_i's edges. Hence $\mathbb{E}[\rho \mid E'_i] = \mathbb{E}[W \mid \hat{y}_{u_i} = z_i] - \mathbb{E}[S \mid E'_i] = 0 - 0$. Changing $\hat{y}_u$ for a non-hub u changes $x_i^\top X_A x_i$ by at most $\sum_{e \ni u, u_i} h_e \le L_u(x_i)$ and changes ψ_u by exactly $|\Phi_u(x_i)| \le L_u(x_i)$ (Lemma A.3); changing a hub $\hat{y}_{u_j}$ ($j \ne i$) changes neither $x_i^\top X_A x_i$ nor any term above, so its Lipschitz constant is 0. McDiarmid with constants $2L_u(x_i)$ gives $\Pr\{|\rho| > \sqrt{2\Lambda^2 \log(2/\eta_2)} \mid E'_i\} \le \eta_2$. Combining by the reverse triangle inequality $|D + S + \rho| \ge |D| - |S| - |\rho|$: on E'_i, with conditional probability $\ge 1 - \eta - \eta_2$,

$$\widetilde{\beta}_A \;\ge\; |x_i^\top X_A x_i| \;\ge\; (1 - m_i)B - \sqrt{2V^- \log(2/\eta)} - \tfrac{\phi^-_{\max}}{3}\log(2/\eta) - \sqrt{2\Lambda^2 \log(2/\eta_2)}.$$

The witness family $\{x_i\}$ is fixed before sampling; only the realized first block is used, through the bound above. Summing the failure probabilities over the disjoint E'_i, $\sum_i \Pr(E'_i)(\eta + \eta_2) \le \eta + \eta_2$, and adding the B event (matrix Bernstein, independent summands $(b_e - p_e)H_e$ with $\|\cdot\|_2 \le 1$ a.s.) gives the claim. □

Proof of Corollary A.5. $Z = x^\top X_A x$ is centered with $\mathrm{Var}(Z) \ge V_1(x) \ge m(1-m)B^2$ (Theorem A.1(ii)) and $|Z| \le Z_{\max}$ surely. By Paley–Zygmund applied to Z^2, $\Pr\{Z^2 \ge \theta \mathbb{E}Z^2\} \ge (1-\theta)^2(\mathbb{E}Z^2)^2/\mathbb{E}Z^4$, and since $\mathbb{E}Z^4 \le Z^2_{\max}\mathbb{E}Z^2$, the ratio is at least $(1-\theta)^2\,\mathbb{E}Z^2/Z^2_{\max} \ge (1-\theta)^2\, m(1-m)B^2/Z^2_{\max}$. □

Proof of Theorem A.6. (i) is the definition of ϕ. (ii) On a triple $e = \{u, w, w'\}$, flatness of u reads $P[w,1]P[w',1] = P[w,0]P[w',0]$; with $P[w,1] = a$, $P[w',1] = 1 - a$ both sides equal $a(1-a)$. (For $r_e \ge 4$, cancellation also depends on the probability product of the remaining members.) (iii) Flatness for every member imposes $\prod_{v \ne u} P[v,1] = \prod_{v \ne u} P[v,0]$ for all $u \in e$. Under positivity, dividing the identity for u by that for u' yields $P[u',1]P[u,0] = P[u',0]P[u,1]$, i.e. a common odds ratio $o = P[v,1]/P[v,0]$ across members; substituting back, $o^{r_e - 1}\prod_{v \ne u} P[v,0] = \prod_{v \ne u} P[v,0]$ with a positive product forces $o^{r_e - 1} = 1$, hence $o = 1$: uniform laws. (iv) For a single-shared pair with shared node u flat on both edges, $\mathbb{E}[a_e a_f \mid \hat{y}_u] = g_{e,u}(\hat{y}_u)g_{f,u}(\hat{y}_u) = p_e p_f$; pairs sharing no node are independent; summing the variance expansion of Proposition A.2 at any x leaves only the diagonal, which equals $\mathrm{Var}(x^\top X_B x)$. □

Appendix D. Mixed-Edge Budget Certificate

For environments with unbounded co-occurrence at a star core, the budget form below complements Theorem A.4. Binary labels, i.i.d. flips; $E = E_* \cup F$ with E_* the disjoint-leaf homogeneous star cores and F arbitrary environment edges not incident to any candidate center. With per-edge rates q_e / π_e (homogeneous / mixed), $\bar{w}_e, \hat{w}_e$ their positive centered magnitudes, and overlaps o_e against a star support Ω_v,

$$\mu_v = \sum_{o_e > 0} \bar{w}_e \frac{o_e}{k_e}, \qquad \tilde{Q}_v = \sum_u \Big(\sum_{e \ni u, o_e > 0} \hat{w}_e \frac{o_e}{k_e} \Big)^2, \qquad \Gamma_v = \mu_v + \sqrt{\tilde{Q}_v \log(1/\eta_2)/2},$$

the A arm obeys $\widetilde{\beta}_A \ge L_A - \Gamma_{v_*}$ with $L_A = (dc_{r,p} - \sqrt{\frac{d}{2}\log(1/\eta)})/r - \bar{\kappa}$ on the first-wrong-center event (conditional probability $\ge 1 - \eta - \eta_2$), and if $\min_v(L_A - \Gamma_v) > U_B$ then $\Pr\{\widetilde{\beta}_A > \widetilde{\beta}_B\} \ge$

$1-(1-p)^t-\eta-\eta_2-\delta_B$. The center-local refinement replaces o_e by the explicit witness entries $b_e = 2\mathbf{1}\{v \in e\} + 2o_e^2/S_{\min}^2$ under the standing assumptions $S_{\min} = d(1-p^{r-1}) - \sqrt{\frac{d}{2}\log(1/\eta)} > dq_r$ and, for the comparison $\Gamma_v^{\mathrm{loc}} \le \Gamma_v$, $2o_e \le S_{\min}^2$; both hold with wide margin in the tested configurations. Given $W_v = 1$ with flipped class c, an incident mixed edge with n_c further members of true label c is inserted with conditional probability $(1-p)^{n_c}p^{k_e-1-n_c}$ (e.g. $(0,1,1)$ at $p = 0.1$: 0.81), which is why center-incident environment edges must be excluded structurally: their conditional rates differ from the unconditional $\bar{w}_e$. On binarized Cora and DBLP structures, observed separation frequency is 99–100% across 100 draws. Native star cores are scarce (Cora: one candidate at $d = 20$, margin $3.4 - 14.3 < 0$ under the center-isolated budget; DBLP: none), and the implanted-core check identifies a conservative Cauchy–Schwarz budget ($\Gamma \approx 98$ versus measured positive overlap mass 0.058), with the local budget $\Gamma^{\mathrm{loc}} \le 0.1$ triggering certification at $d \ge 100$ ($p = 0.05$) and $d \ge 150$ ($p = 0.10$) on both datasets. Full proofs are retained in the repository record.

Appendix E. Conditioning Boundary

For a shared training state Θ, the conditional variance identity holds only under conditional independence of node predictions given Θ; total variance then separates within-state from across-state components. The real-predictor experiments condition on fixed features and vary training partitions or seed labels. They measure centered amplification under training-induced prediction variability; fresh-feature variability requires a separate sampling design.

Appendix F. Conditional Risk Calculation and Native Scope Checks

Exact conditional moments and tail bound. For $g_e(z) = \Pr(a_e = 1 \mid \hat{y}_U = z)$, conflicting pinned labels give $g_e = 0$; agreeing pinned labels c give $g_e = \prod_{u \in e \setminus U} P[u, c]$; an edge not meeting U has $g_e = p_e$. Thus $M_U(z) = \sum_e (g_e - p_e)H_e$. For $h_e(q_z) = (\sum_{u \in e} q_{z,u})^2/r_e$, the directional variance is

$$v_z = \sum_{e,f} \left[\Pr(a_e = a_f = 1 \mid z) - g_e(z)g_f(z)\right] h_e(q_z)h_f(q_z).$$

The joint probability enforces agreement within each edge, respecting pinned labels; a common label across the union is required only when the unpinned sets overlap. Disjoint unpinned sets have zero conditional covariance; remaining shared nodes and duplicate slots are retained. With $Y_z = \mathrm{sgn}(m_z)q_z^\top X_A q_z$, $\mathbb{E}[Y_z \mid z] = |m_z|$. For $D_z > 0$, Cantelli bounds $\Pr(Y_z \le \tau s_z \mid z) \le v_z/(v_z + D_z^2)$, and $Y_z > \tau s_z$ implies $\|X_A\|_2 > \tau$. Summing over disjoint states proves Eq. (A.1); zero variance gives bound one when $D_z > 0$. Finite decimal model probabilities and saved dyadic directions are evaluated as exact rationals; an approximate eigenvector is a valid fixed witness. An identically zero mean uses a fixed coordinate witness.

Finite calibration and confirmation. For nominal tail $\alpha = 0.05$, the threshold is the 1,929th ordered certified upper norm enclosure among 2,000 independent B draws. The exact binomial probability $\Pr\{\mathrm{Bin}(2000, 0.95) \ge 1929\} \le 1/800$ bounds the chance that its true B tail exceeds 5%; a union bound over eight cells gives at least 99% joint calibration confidence. Strict exceedance and upper enclosures make ties and numerical uncertainty conservative. Each validation realization consists of a complete node-prediction draw. Eight-cell adjusted exact binomial intervals retain ambiguous counts at both endpoints. The coordinate-only certificate was inconclusive. After developing group-based witnesses, we froze directions and exact moments before the new calibration and validation streams; threshold-dependent bounds were saved before validation. Protocols, raw intervals, and the unsuccessful original experiment remain in `notes/native-risk/`, indexed by the README.

Earlier native variance diagnostics. Under binary majority-versus-rest flips at $p = 0.05/0.15$, the maximum-degree witness is chosen from structure only. Cora's witness (degree 145) carries 94.7% of V_1 at its coordinate direction; DBLP's top two nodes share the same edges and carry 98%. In the original 30-draw probe, mean A/B norms were 7.88/2.62 and 12.57/3.85 on Cora, and 5.85/2.40 and 5.96/2.74 on DBLP. The exact independent-edge directional variances are 2.04/4.15 and 0.27/0.56, with first-order lower-bound ratios $V_1/\operatorname{Var}_B = 41.3/31.5$ and 12.0/8.8, respectively. These comparisons use exact directional variances and first-order quantities. Theorem A.4's concentration penalties leave the native norm-separation certificate inactive: $\Lambda^2 = 128.5$ on Cora and 45.6 on DBLP leave insufficient drift. The mixed-edge budget also fails to certify a native star core. Conditional risk analysis provides a separate tail-risk certificate using exact group-conditioned moments.

Fixed spectral-subspace readout. The registered readout compares each realization's top-k eigenspace ($k = 8$ fixed) with that of the common expected operator $\sum_e p_e H_e$, using projector distance $\|QQ^\top - Q_{\mathrm{ref}}Q_{\mathrm{ref}}^\top\|_F$. Table F.1 reports paired draw-level differences and reference eigengaps for 30 draws per setting. Cora's intervals exclude zero at both noise levels; DBLP's include zero. At $p = 0.15$ its positive but small gap may make the unique reference subspace perturbation-sensitive. The readout uses the registered value $k = 8$ throughout.

Dataset	p	Gap	Mean A	Mean B	A−B [95% CI]
Cora	0.05	0.582	1.474	0.943	$+0.531$ $[+0.292, +0.781]$
Cora	0.15	0.641	2.075	1.502	$+0.573$ $[+0.373, +0.792]$
DBLP	0.05	0.589	1.394	1.642	-0.247 $[-0.648, +0.167]$
DBLP	0.15	0.017	2.475	2.524	-0.049 $[-0.266, +0.161]$

Table F.1: Fixed-k projector-distance readout against the common expected operator. Means report projector distances; intervals describe paired draw-level differences on the fixed native structures.

Appendix G. Experimental Protocol and Sensitivity

Prototype protocol. Cora-CC: 2708 nodes, 1579 edges, 7 classes; DBLP-CA: 41,302 nodes, 22,363 edges, 6 classes. Five-fold cross-fitting; calibration seeds 31000+, evaluation seeds 32000+, nested calibration sizes $\{20, 40, 80\}$; every bootstrap replicate resamples calibration runs, recomputes $\hat{q}_e$ and $\hat{M}$, regenerates 20 B draws, and resamples the 20 evaluation runs (1000 replicates; bootstrap seeds $99+N_{\mathrm{cal}}$ per calibration size). All twelve fixed-split intervals have lower endpoints above one (range 1.165–3.274). Three fully disjoint outer batches (seeds 33/36k, 41/44k, 49/52k blocks, disjoint from all other blocks) repeat the protocol; all 24 batch intervals have lower endpoints above one (smallest 1.016, DBLP-tri batch 2), so ratio magnitudes vary across splits while sign evidence is stable. Spectral norms in the batch runs use a 25-step reorthogonalized Lanczos estimator validated against `eigsh` (worst relative error $< 10^{-7}$, recorded per cell).

GCN and label propagation. Cora GCN: row-normalized features, two layers of width 64, 500 epochs, lr 0.2, five-fold cross-fitting on the symmetric-normalized clique expansion (a width-16 configuration collapsed to one class and is excluded). LP: 20% seeded true labels, 50 iterations, propagation 0.9, restart 0.1. Walmart-trips: induced subgraph on the first 20,000 original nodes (19,965 incident nodes, 18,034 edges, 11 categories, sizes 2–25; no node features, LP only). A DBLP-GCN configuration exists in the saved record (ratio 2.15) at near-chance accuracy (0.31) and is not tabled.

Controlled-family and native protocol. Family A uses two-sided concentration constants and realized co-occurrences, first certifying at $d = 250$. The 30-draw samples describe observed norm fluctuations. The structure and native experiments run under the frozen protocol `upgrade/protocol.json` (hash recorded in each result file); pilot records (64 draws) are kept separate from formal intervals (256 structure / 30 native draws); seeds are disjoint from all earlier blocks; every registered unit including solver failures is recorded. Fresh-seed reproductions and the six canonical exhaustive checks are indexed in the accompanying README.

Homogeneous-star and cyclic verification. Tables G.1 and G.2 give the star and cyclic grids ($r = 3$, $t = 100$). The star guarantee at $p = 0.05$, $d = 100$ is ≥ 0.974; empirical grid separation is 99.2–100%.

p	d	L_A	U_B	no center	A/B fail	sep. fail [95% CI]
0.05	20	3.44	12.72	34/5000	0/0	31 [.0044, .0088]
0.05	50	10.67	16.14	38/5000	0/0	38 [.0055, .0104]
0.05	100	23.44	19.93	29/5000	0/0	19 [.0024, .0059]
0.05	200	49.85	25.26	31/5000	0/0	6 [.0006, .0026]
0.10	20	2.54	12.90	0/5000	0/0	0 [.0000, .0008]
0.10	50	8.42	17.03	0/5000	0/0	0 [.0000, .0008]
0.10	100	18.94	21.68	0/5000	0/0	0 [.0000, .0008]
0.10	200	40.85	28.28	0/5000	0/0	0 [.0000, .0008]

Table G.1: Joint Monte Carlo on disjoint stars (5000 draws/cell, $\eta = \delta_B = 0.01$). "no center": no wrong center; "A/B fail": $\widetilde{\beta}_A < L_A$ (given a wrong center) and $\widetilde{\beta}_B > U_B$; allowed separation-failure rates are 0.0259 ($p = 0.05$) and 0.0200 ($p = 0.10$).

d	Q	$\|\Sigma H_e\|_2$	L_A^{ov}	U_B	Theory	Empirical	$\overline{\widetilde{\beta}}_A$	$\overline{\widetilde{\beta}}_B$
20	78	8.0	1.23	12.43	×	100%	6.80	2.39
50	198	18.0	7.13	15.71	×	100%	15.35	3.32
100	398	34.7	18.41	19.39	×	99.8%	29.52	4.34
150	598	51.3	30.38	22.21	✓	99.8%	43.63	5.15
200	798	68.0	42.71	24.59	✓	100%	57.79	5.79

Table G.2: Cyclic overlap verification (r=3, t=100, p=0.05, $\eta = \delta_B = 0.01$, 500 draws per d, seed 42). Structural quantities match $4d-2$ and $(d+4)/3$ exactly.

Appendix H. Supporting Normalized Diagnostics

H.1. Local response and a perturbation tool

At a deterministic baseline $A_0 = B(p) + \kappa I$ with degree matrix D_0, perturb by $E = A - A_0$ and put $C = D_0^{-1}\,\mathrm{diag}(E\mathbf{1})$. Differentiation gives

$$J_{A_0}(E) = D_0^{-1/2} E D_0^{-1/2} - \tfrac{1}{2}\{CS(p) + S(p)C\}.$$

The two terms describe relation change and degree correction at the same baseline. They are not an expansion around an arm-specific expectation.

Lemma H.1 (Positive-degree perturbation bound)**.** *Let A, A' be symmetric nonnegative matrices, with row-sum diagonal matrices D, D' and minimum degree $d_0 > 0$ across both. Then*

$$\|D^{-1/2} A D^{-1/2} - D'^{-1/2} A' D'^{-1/2}\|_2 \leq \frac{\|A - A'\|_2}{d_0} + \frac{\|A'\|_2 \|D - D'\|_2}{d_0^2}.$$

Proof. Insert $D^{-1/2}A'D^{-1/2}$ and split the remaining normalization difference into a left and a right factor. The scalar mean-value bound $|x^{-1/2} - y^{-1/2}| \le |x - y|/(2d_0^{3/2})$ applies to each diagonal entry. Each of the latter two terms is bounded by $\|A'\|_2\|D - D'\|_2/(2d_0^2)$; the first is bounded by $\|A - A'\|_2/d_0$. □

For a node-u replacement, $|d_v - d'_v|$ is bounded by the number of edge slots containing both u and v. Including $v = u$ makes the maximum of these counts equal to $\deg(u)$. The hub factor must remain in the bound. Since $d_0 \ge \kappa$, replacing d_0 by κ is conservative, not an assertion of equality.

H.2. Visual calibration protocol and conditional-risk analysis

The visual objects use the ED-HNN preprocessing [16] of NTU2012 and ModelNet40. Input SHA-256 values are `f59d6f62...30d2c` and `3de34493...ee8b8` for NTU2012 content and edges, and `51565c05...8604` and `e6e0ae37...4048` for ModelNet40. The fixed stratified split contains 1,182/377/453 NTU objects and 7,372/2,446/2,493 ModelNet objects (training/validation/test). The predictor uses hidden width 64, Adam with learning rate 0.01 and weight decay 5×10^{-4}, full-batch training and validation-loss checkpointing. The epoch ceiling is 100 on NTU and 400 on ModelNet, selected using ten development runs per dataset. Run seeds are `2026102100 + dataset`$\cdot 10^5$ `+ batch`$\cdot 10^4$ `+ role`$\cdot 10^3$ `+ index`; reference, calibration, evaluation and development streams are disjoint. One batch per dataset contains 30/99/200 reference/calibration/evaluation complete runs. The reference center is shared by all three calibrators and remains frozen throughout their comparison.

Full counts, thresholds, tie rates, spectral residuals, timings and input hashes are retained in the reproducibility materials under `experiments/spectral-risk-calibration/results/visual_alarm_v1/`. No spectral solver fallback occurred in the 658 complete training runs. These caches preserve the distinction between independent training runs and simulated draws from a fitted node or edge law.

Marginal and conditional probability targets. For a fixed reference center, let $\mathcal{C}$ denote the random calibration sample and write $r(\mathcal{C}) = \Pr_{\mathcal{L}}\{T_{\text{new}} > q(\mathcal{C}) \mid \text{reference}, \mathcal{C}\}$. Proposition 4.1 bounds the average over calibration samples, $\mathbb{E}_{\mathcal{C}} r(\mathcal{C}) \le \alpha$. A binomial tolerance rule instead bounds the calibration-sample probability that $r(\mathcal{C})$ exceeds α.

The NTU frozen threshold gave 28/200 exceedances. With its reference center fixed, 10,000 random 99/200 splits of the existing 299 scores gave a mean of 5.0%, median 4.5%, and central 95% range 1.0–11.0%; 4.4% of splits exceeded 10%. The frozen threshold lay at the 0.4th percentile of resampled thresholds. ModelNet gave mean 5.0%, median 4.5%, and central 95% range 1.0–11.0%. These averages describe repeated random partitions of one finite score pool. Their agreement with the nominal target is an internal rank-consistency check, not a new independent test of the population guarantee. Uniform rounding of both score sets to 6–14 decimals changed the NTU count from 28 to 27 (13.5%), so near ties account for 0.5 percentage points of its observed rate.

Binomial tolerance calibration. For an independent-and-identically-distributed run mechanism, conditional on the frozen reference stage, choose

$$k_{\alpha,\delta} = \min\{k \in \{1, \dots, M\} : \Pr\{\text{Binomial}(M, 1 - \alpha) \ge k\} \le \delta\}. \tag{H.1}$$

If no finite index satisfies the criterion, use an infinite threshold. With $q = T_{(k_{\alpha,\delta})}$ and strict exceedance, the established tolerance bound is

$$\Pr_{\mathcal{C}}\{r(\mathcal{C}) > \alpha \mid \text{reference}\} \le \delta$$

[12]. This high-probability statement requires the independent-run population model, rather than only the finite exchangeability used for the marginal rank argument.

At $\alpha = \delta = .05$, the smallest admissible index is 98 for $M = 99$ (binomial tail 0.039), 195 for $M = 199$ (0.027), 483 for $M = 499$ (0.035), and 961 for $M = 999$ (0.044). At $M = 99$, the marginal rule instead takes index 95. Evaluating the tolerance threshold on the same observed evaluation runs gives 19/200 (9.5%) on NTU and 3/200 (1.5%) on ModelNet. This comparison changes the probability target and threshold, not the original frozen observations. Because the comparison was chosen after observing the original evaluation results, it does not supply an independent validation of the new rule. Prospective use fixes α, δ, M and the selection criterion before new calibration and evaluation runs are generated.

H.3. Secondary response to feature perturbations

Each evaluation model was re-inferred on five paired feature-noise strengths $\sigma \in \{0, .1, .25, .5, 1\}$. Only non-training object features were perturbed, with one Gaussian noise field shared across strengths within each run; the candidate hypergraph, preprocessing, model weights and reference center remained fixed. These experiments compare score responses to this specified perturbation mechanism, separately from the cross-domain calibration comparison in Section 5.4.

At $\sigma = 1$ and a retrospective realized 10% false-positive rate, T detection was 65.5% on NTU versus 92.5% for Frobenius deviation (AUC 0.879 versus 0.976); on ModelNet it was 66.5% versus 100% (AUC 0.925 versus 0.998), with predictive entropy at 100% (AUC 1.000). Observed T-minus-Frobenius differences were -27.0 and -33.5 percentage points, with paired-run percentile bootstrap 95% intervals $[-36.5, -18.5]$ and $[-41.0, -24.0]$, respectively. The curves use the same evaluation scores for threshold selection and comparison, so they describe in-sample ranking rather than calibrated deployment performance.

The two operator norms measure different functionals. For the symmetric deviation $\Delta = S - \widehat{M}$,

$$\|\Delta\|_2 = \max_i |\lambda_i(\Delta)|, \qquad \|\Delta\|_F = \sqrt{\sum_i \lambda_i(\Delta)^2}.$$

An increase spread across several spectral directions can therefore raise Frobenius deviation more than the largest directional component. This offers a possible explanation for the stronger aggregate response under the tested Gaussian feature perturbations; nonlinear prediction and normalization do not themselves imply an isotropic operator perturbation. These measurements do not establish an advantage for T under localized or coherent changes, which would require a separate comparison.

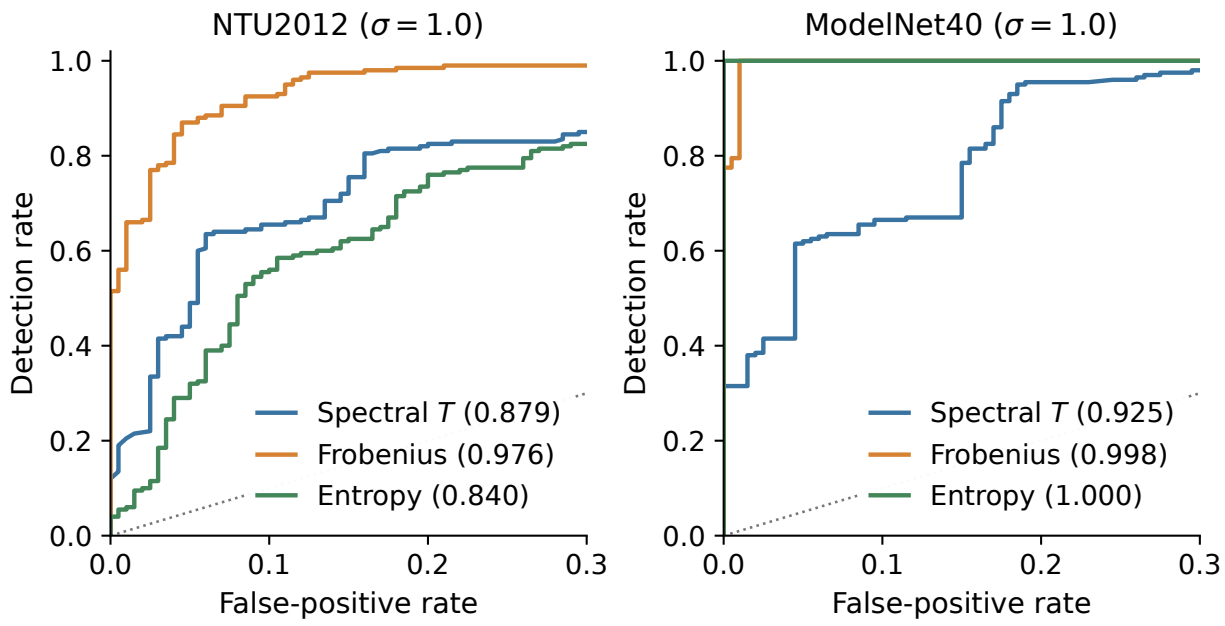


Figure H.1: Secondary feature-perturbation comparison at $\sigma = 1$. Clean evaluation scores are negatives and paired perturbed scores are positives. Thresholds are swept on the same evaluation data, so these ROC curves describe score ranking rather than calibrated deployment rules. The displayed interval is the low-false-positive-rate portion of each curve; AUC values refer to the full curve.